

# Module 1: Basic Linear Algebra

## *Previous Year University Examination Questions (Model Collection)*

### **PART A – Short Answer Questions (2–5 Marks)**

- 1. Define a matrix. Give examples of square, identity, and zero matrices.
- 2. Write the system of linear equations in the matrix form  $Ax = b$ .
- 3. Define homogeneous and non-homogeneous systems of linear equations.
- 4. What is a trivial solution?
- 5. Define linear independence and linear dependence.
- 6. State the condition for a set of vectors to form a basis.
- 7. Define rank of a matrix. How is it found using row-echelon form?
- 8. Define kernel and image of a linear transformation.

### **PART B – Problem Solving Questions (5–8 Marks)**

- 9. Find the rank of the matrix using row reduction method:  $\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$ .
- 10. Determine whether the vectors  $(1,0,1)$ ,  $(2,1,3)$ ,  $(3,1,4)$  are linearly independent.
- 11. Verify whether the set  $\{(1,0,0), (0,1,0), (0,0,1)\}$  forms a basis for  $\mathbb{R}^3$ .
- 12. Solve the homogeneous system  $Ax = 0$  and find its trivial solution.
- 13. For a  $3 \times 3$  matrix  $A$ , if  $\det(A) = 0$ , what can you say about its rank and solutions?
- 14. Find the nullity of a matrix if  $\text{rank}(A) = 2$  and dimension = 3.
- 15. Find the transformation matrix for  $T: \mathbb{R}^2 \rightarrow \mathbb{R}^3$ .

### **PART C – Long Answer / Theory Questions (8–12 Marks)**

- 16. State and prove the Rank–Nullity Theorem.
- 17. Prove that if determinant of a matrix is non-zero, then the matrix is invertible.
- 18. Show that a set containing more vectors than the dimension of the space is linearly dependent.
- 19. Prove that the standard basis vectors form a basis for  $\mathbb{R}^n$ .

- 20. Discuss the relationship between kernel, image, rank, and nullity of a linear transformation.
- 21. Determine whether a given system of linear equations is consistent using rank method.